

Toolkit Documentation – Brain Atlases and ROI Collection

This toolkit compiles a variety of publicly available brain atlases and regions of interest (ROIs) derived from them, intended for reference-region analysis in neuroimaging or general-purpose ROI analysis. **All data are restricted to non-commercial research use.** Please carefully review the copyright and citation requirements for each section before use.

Usage Restrictions: None of the atlases or ROIs included in this toolkit may be used for commercially sponsored research, clinical diagnosis, treatment, or patient care. If multiple atlases are combined within a single project, please ensure that the citation and licensing requirements of each atlas are satisfied independently.

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1. Desikan-Killiany Atlas

Atlas Description

A cortical parcellation atlas based on surface morphology that divides the cerebral cortex into gyrus-based regions of interest. This toolkit uses the SPM12-compatible MNI-space volumetric version (`aparc+aseg` , containing **175 original ROIs**), derived from the FreeSurfer `cvs_avg35_inMNI152` template and including both cortical parcellations and subcortical structure labels.

- **Source:** Harvard Center for Morphometric Analysis / FreeSurfer software suite
- **Space:** MNI space (FreeSurfer `cvs_avg35_inMNI152` template)
- **Official page:** <https://surfer.nmr.mgh.harvard.edu/fswiki/CorticalParcellation>

Custom Merged ROIs

The following three ROI masks were extracted and merged from the original atlas:

ROI Name	Original Indices	Description
Desikan_Killiany_MNI_Brain_Stem	16	Brainstem
Desikan_Killiany_MNI_Cerebellum_Cortex	8, 47	Bilateral cerebellar cortex (white matter excluded)
Desikan_Killiany_MNI_Cerebellum	7, 8, 46, 47	Whole cerebellum (bilateral white matter and cortex)

Copyright and Citation

Desikan RS, Ségonne F, Fischl B, et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage*. 2006;31(3):968-980. doi:10.1016/j.neuroimage.2006.01.021

2. Neuromorphometrics Multi-Atlas Label Set

Atlas Description

A maximum-probability tissue label set derived from the **MICCAI 2012 Multi-Atlas Labelling Challenge**. The original MRI scans come from the OASIS project, and all images have been registered to MNI space.

- **Source:** MICCAI 2012 Challenge dataset; Neuromorphometrics, Inc. academic subscription edition
- **Space:** MNI space
- **Official page:** <http://Neuromorphometrics.com/>

Custom Merged ROIs

The following ROIs are all constructed from Neuromorphometrics label indices:

ROI Name	Component Indices	Description
Neuromorphometrics_BrainStem	35	Brainstem
Neuromorphometrics_Cerebellum	38, 39, 40, 41, 71, 72, 73	Whole cerebellum (gray matter, white matter, vermis)
Neuromorphometrics_CerebellumGray	38, 39, 71, 72, 73	Cerebellar gray matter (white matter excluded)
Neuromorphometrics_CerebellumWhite	40, 41	Cerebellar white matter
Neuromorphometrics_CerebellumBrainStem	35, 38, 39, 40, 41, 71, 72, 73	Brainstem + whole cerebellum
Neuromorphometrics_CerebralWhiteMatter	44, 45	Cerebral white matter

Copyright and Citation

All publications that use these data must include the following acknowledgements:

- MRI scans are derived from the **OASIS project** (<http://www.oasis-brains.org/>)
- Labeled data are provided by **Neuromorphometrics, Inc.** under academic subscription (<http://Neuromorphometrics.com/>)

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3. AAL3 Template

Atlas Description

Automated Anatomical Labelling Atlas, version 3 (AAL3).

The file used in this toolkit is `AAL3v1_1mm.nii` (1×1×1 mm resolution) from the **AAL3v1_for_SPM12** package, containing **166 ROIs**, with a maximum label value of 170.

- Source: AAL3v1_for_SPM12 (<https://www.oxcns.org/aal3.html>)
- Space: MNI space, 1×1×1 mm
- Official page: <https://www.oxcns.org/aal3.html>

Custom Merged ROIs

The following ROI is generated by merging regions from `AAL3v1_1mm.nii` :

ROI Name	Included Original ROIs	Original Indices	Description
<code>AAL3_Cerebellum_Inferior</code>	Cerebelum_Crus2_L, Cerebelum_Crus2_R, Cerebelum_7b_L, Cerebelum_7b_R, Cerebelum_8_L, Cerebelum_8_R, Cerebelum_9_L, Cerebelum_9_R, Cerebelum_10_L, Cerebelum_10_R	97, 98, 105, 106, 107, 108, 109, 110, 111, 112	Inferior cerebellar gray matter (Crus II, lobules VIIb –X, bilateral)

Copyright and Citation

Rolls ET, Huang CC, Lin CP, Feng J, Joliot M. Automated anatomical labelling atlas 3. *Neuroimage*. 2020;206:116189. doi:10.1016/j.neuroimage.2019.116189

4. Talairach Daemon (TD) Atlas Series

Atlas Description

The Talairach Daemon (TD) atlas series originates from the **Talairach Daemon database**, which digitizes the 1988 Talairach atlas into a queryable hierarchical anatomical labeling system (spanning hemisphere, lobe, sub-lobe, gyrus, nucleus, etc.).

All TD atlas files included in this toolkit are taken directly from the `MNI_atlas_templates` directory of the **WFU PickAtlas v3.0** toolbox. The WFU PickAtlas development team has already performed the transformation from Talairach space to MNI space (using the nonlinear tal2mni transformation described by Matthew Brett, with 0.9-coefficient layer-wise extrapolation applied to fill the apex regions). The atlas files in this directory are **already in MNI space**, with image dimensions of 91×109×91 voxels and 2 mm sampling.

This toolkit includes the following four files from the TD series:

Filename	Atlas Contents	ROI Count
TD_brodmann.nii	Brodmann areas + subcortical nuclei	75
TD_hemisphere.nii	Hemisphere divisions (L/R cerebrum, cerebellum, brainstem, etc.)	7
TD_label.nii	Gyrus anatomical labels	55
TD_lobe.nii	Lobe divisions	12

- **File source:** WFU PickAtlas v3.0, MNI_atlas_templates/ directory
- **Original database:** Talairach Daemon, developed by the Research Imaging Center, University of Texas Health Science Center at San Antonio
- **MNI conversion method:** Matthew Brett's tal2mni nonlinear transformation, performed by the WFU PickAtlas team

Custom Merged ROIs

The occipital lobe is extracted from TD_lobe.nii as a standalone reference region:

ROI Name	Source Atlas	Original ROI	Original Index	Description
TD_lobe_Occipital_Lobe	TD_lobe.nii	Occipital Lobe	9	Occipital lobe (MNI space)

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WFU PickAtlas toolbox (source of the MNI conversion):

Maldjian JA, Laurienti PJ, Kraft RA, Burdette JH. An automated method for neuroanatomic and cytoarchitectonic atlas-based interrogation of fMRI data sets. *Neuroimage*. 2003;19(3):1233-1239. doi:10.1016/s1053-8119(03)00169-1

Maldjian JA, Laurienti PJ, Burdette JH. Precentral gyrus discrepancy in electronic versions of the Talairach atlas. *Neuroimage*. 2004;21(1):450-455. doi:10.1016/j.neuroimage.2003.09.032

Talairach Daemon database:

Lancaster JL, Rainey LH, Summerlin JL, et al. Automated labeling of the human brain: a preliminary report on the development and evaluation of a forward-transform method. *Hum Brain Mapp*. 1997;5(4):238-242. doi:10.1002/(SICI)1097-0193(1997)5:4<238::AID-HBM6>3.0.CO;2-4

Original Talairach atlas:

Talairach J, Tournoux P. (1988) *Co-planar stereotaxic atlas of the human brain*. Thieme, New York.

Copyright notice: The Talairach Daemon is copyright of the Research Imaging Center, University of Texas Health Science Center at San Antonio. WFU PickAtlas is copyright of the Wake Forest University ANSIR Laboratory.

5. SPM12 Tissue Probability Map (TPM) Derived Masks

Atlas Description

The following masks are generated from the built-in files in the `spm12/tpm/` directory of a standard SPM12 installation. They reside in MNI space and are consistent with the standard SPM12 templates.

- **Source:** Standard SPM12 installation, `spm12/tpm/` directory
- **Space:** MNI space

Derived ROIs and Masks

White matter probability masks (source: frame 2 of `TPM.nii`, i.e., the white matter probability map):

ROI Name	Threshold	Description
<code>WhiteMatter</code>	White matter probability > 0.5	Combined cerebral + cerebellar white matter region
<code>whitemask095</code>	White matter probability > 0.95	High-confidence white matter region

Whole-brain intracranial mask (source: `mask_ICV.nii`):

ROI Name	Description
mask_ICV	Intracranial volume (ICV) mask, usable for ICV estimation or whole-brain analysis

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Friston, K.J., Ashburner, J.T., Kiebel, S.J., Nichols, T.E., Penny, W.D. (Eds.) (2007) *Statistical Parametric Mapping: The Analysis of Functional Brain Images*. Academic Press (Elsevier), London.

License: Subject to the non-commercial research license of the SPM12 software.

6. LPBA40 Atlas

Atlas Description

The LONI Probabilistic Brain Atlas (LPBA40) is constructed from whole-brain MRI data of 40 healthy volunteers. Each subject's MRI was manually segmented, yielding annotations for **56 brain structures** (primarily cortical). All annotations were then transformed into a common atlas space to generate a maximum probability map.

The version used in this toolkit has been registered to **MNI152NLin2009cAsym space** and post-processed to fill residual holes (median-value filling) and apply a brain mask to remove non-brain regions.

- **Source:** LONI / UCLA Brain Mapping Center
- **Space:** MNI152NLin2009cAsym
- **Official page:** <http://www.loni.usc.edu/research/atlasses>

Custom Merged ROIs

(No merged ROIs based on LPBA40 are included in the current version of this toolkit.)

Copyright and Citation

Shattuck DW, Mirza M, Adisetiyo V, et al. Construction of a 3D probabilistic atlas of human cortical structures. *Neuroimage*. 2008;39(3):1064-1080.
doi:10.1016/j.neuroimage.2007.09.031

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